

Deterministic Optimal Control

Pontryagin's Minimum Principle and LQR

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Deterministic Optimal Control

Continuous-time problem:

$$\min_{u(\cdot)} \int_0^T \ell(x(t), u(t)) dt + \ell_f(x(T))$$

$$\text{s.t. } \dot{x}(t) = f(x(t), u(t))$$

Infinite-dimensional optimization problem.

Solution: Open-loop trajectory.

Discrete-Time Formulation

After discretization:

$$\min_{\{u_k\}} \sum_{k=0}^{N-1} \ell(x_k, u_k) + \ell_f(x_N)$$

$$x_{k+1} = f(x_k, u_k)$$

$$u_{\min} \leq u_k \leq u_{\max}$$

Now finite-dimensional nonlinear program.

State and control samples = **knot points**.

This is simply:

$$\min_{x_0, \dots, x_N, u_0, \dots, u_{N-1}} J$$

$$\text{s.t. } x_{k+1} - f(x_k, u_k) = 0$$

Large equality-constrained optimization.

Therefore:

Pontryagin = KKT conditions for this problem.

Forming the Lagrangian

Introduce multipliers:

$$\lambda_{k+1}$$

Lagrangian:

$$\mathcal{L} = \sum_{k=0}^{N-1} \ell(x_k, u_k) + \ell_f(x_N) + \sum_{k=0}^{N-1} \lambda_{k+1}^T (f(x_k, u_k) - x_{k+1})$$

Hamiltonian Definition

Define discrete-time Hamiltonian:

$$H(x_k, u_k, \lambda_{k+1}) = \ell(x_k, u_k) + \lambda_{k+1}^T f(x_k, u_k)$$

Then:

$$\mathcal{L} = \sum_{k=0}^{N-1} \left(H(x_k, u_k, \lambda_{k+1}) - \lambda_{k+1}^T x_{k+1} \right) + \ell_f(x_N)$$

Pontryagin's Minimum Principle

Taking derivatives:

1. State Dynamics

$$x_{k+1} = \nabla_{\lambda} H(x_k, u_k, \lambda_{k+1}) = f(x_k, u_k)$$

2. Costate Dynamics

$$\lambda_k = \nabla_x H(x_k, u_k, \lambda_{k+1})$$

3. Optimal Control

$$u_k = \arg \min_{u \in \mathcal{U}} H(x_k, u, \lambda_{k+1})$$

Costate Equation Explicitly

Compute:

$$\lambda_k = \nabla_x \ell(x_k, u_k) + \nabla_x f(x_k, u_k)^T \lambda_{k+1}$$

Terminal condition:

$$\lambda_N = \nabla_x \ell_f(x_N)$$

We now have forward-backward system:

- Forward: state - Backward: costate

Continuous-Time Version

Continuous-time PMP:

$$\dot{x} = \nabla_{\lambda} H$$

$$\dot{\lambda} = -\nabla_x H$$

$$u = \arg \min H$$

Costate = adjoint variable.

This is a boundary-value problem.

Indirect (Shooting) Methods

Algorithm:

1. Guess control sequence. 2. Forward simulate state. 3. Backward integrate costate. 4. Compute control gradient. 5. Line search. 6. Repeat.

Called:

- Indirect methods - Shooting methods

Linear Quadratic Regulator (LQR)

$$\min \sum_{k=0}^{N-1} x_k^T Q x_k + u_k^T R u_k + x_N^T Q_f x_N$$

$$x_{k+1} = A x_k + B u_k$$

$$Q \succeq 0, \quad R \succ 0$$

Convex, analytic solution exists.

$$H = x_k^T Q x_k + u_k^T R u_k + \lambda_{k+1}^T (A x_k + B u_k)$$

Optimality condition:

$$\nabla_u H = 0$$

$$2R u_k + B^T \lambda_{k+1} = 0$$

$$u_k = -\frac{1}{2} R^{-1} B^T \lambda_{k+1}$$

Costate Equation for LQR

$$\lambda_k = 2Qx_k + A^T \lambda_{k+1}$$

Assume linear relation:

$$\lambda_k = P_k x_k$$

Substitute $\rightarrow$

Leads to Riccati recursion.

Double Integrator Example

State:

$$x = \begin{bmatrix} \textit{position} \\ \textit{velocity} \end{bmatrix}$$

Dynamics:

$$x_{k+1} = \begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ h \end{bmatrix} u_k$$

Interpretation:

Brick sliding on ice (no friction).

Key Insights

1. Optimal control = constrained optimization. 2. Pontryagin = KKT for trajectory problem. 3. Hamiltonian encodes everything. 4. Costate is Lagrange multiplier. 5. LQR gives analytic feedback solution. 6. Shooting integrates forward + backward.

Modern MPC builds on this foundation.